

Project Topic 2: Assess Fitness App Usage & Calorie Burn Patterns to Drive Personalised Engagement (*FitTech*)

Context:

A leading fitness technology platform has curated a comprehensive dataset to analyze **user activity patterns, workout engagement, and app feature usage** across multiple regions. The dataset includes **daily workout logs, wearable monitoring data, subscription and profile records, and in-app engagement sessions**. Your role as a data analyst is to explore and analyze this dataset to help the organization answer critical aspects such as:

Business Objective 1: To identify high-engagement workout types, time slots, and devices to guide program design and personalization.

Sample Business Questions

- Which workout types (Cardio, Yoga, Strength, Mix) drive the highest calorie burn and participation?
- What time of day (Morning vs Evening) shows peak activity across users?

Business Objective 2: To understand retention and subscription performance across demographics, regions, and user goals.

Sample Business Questions

- How do Premium users differ in activity levels and retention compared to Free users?
- What is the average consistency score across age groups and regions?

Business Objective 3: To analyze app feature usage and engagement drivers for improved user experience and product strategy.

Sample Business Questions

- Which features (Workout Tracker, Diet Log, Community, Progress) drive the most in-app activity?
- How does notification click-through rate vary by feature?

Business Objective 4: To leverage performance metrics and personalization insights to optimize workout efficiency and user satisfaction.

Sample Business Questions

- What is the workout efficiency index (Calories ÷ Duration) across different age groups?
- How does device usage (Mobile, Smartwatch, Web) influence workout completion?